

**NAAN MUDHALVAN**

**PROJECT DOCUMENTATION**

**SALESFORCE CRM  
IMPLEMENTATION**

**Courier Management System**

|                     |                                                                          |
|---------------------|--------------------------------------------------------------------------|
| <b>Platform</b>     | Salesforce Lightning Experience                                          |
| <b>Project Type</b> | Courier Management System (CMS)                                          |
| <b>Purpose</b>      | Customer, account, order, automation, reporting and dashboard management |

Submitted as part of the Naan Mudhalvan practical project documentation.

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# 1. INTRODUCTION

## 1.1 Project Overview

Courier businesses handle many activities such as customer registration, parcel booking, branch coordination, delivery assignment, status updates and billing. When these tasks are maintained through separate registers or spreadsheets, information can become difficult to locate and updates may be delayed.

The proposed Courier Management System (CMS) brings these activities into one Salesforce environment. It provides a structured way to store courier information, connect shipments with customers and delivery agents, automate selected processes, and present operational information through reports and dashboards.

The system is designed as a cloud-based academic project using Salesforce objects, relationships, validation rules, page layouts, flows, profiles, roles and reporting features.

## 1.2 Purpose

The purpose of CMS is to provide a centralized platform for recording and monitoring courier operations. The solution focuses on reducing repeated manual work, improving data consistency, supporting shipment visibility and helping staff review operational information.

Main objectives are:

- Maintain customer and shipment information in organized records.
- Connect shipments with delivery agents and branches.
- Use validation rules to reduce incorrect or incomplete entries.
- Automate selected status updates using Salesforce Flow.
- Provide role-based access for different users.
- Create reports and dashboards for operational monitoring.

## 1.3 User Story

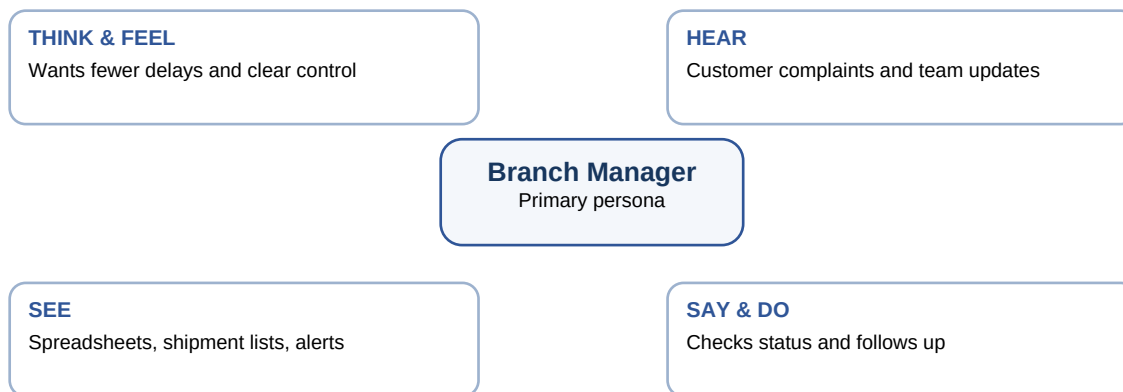
**As a branch manager, I want to manage shipments, agents and delivery updates from one system so that daily courier operations can be monitored without depending on multiple manual records.**

## 2. IDEATION PHASE

### 2.1 Problem Statement

| User           | Goal                         | Current Difficulty                 | Effect                     |
|----------------|------------------------------|------------------------------------|----------------------------|
| Branch Manager | Manage shipments efficiently | Tracking is spread across records  | More follow-up work        |
| Customer       | Know parcel status           | Updates may not be centralized     | Uncertainty about delivery |
| Delivery Agent | Receive clear assignments    | Information may be shared manually | Risk of missed updates     |
| Support Staff  | Answer tracking queries      | Need to check multiple records     | Slower responses           |

### 2.2 Empathy Map



The branch manager is treated as the primary persona because this role needs visibility across shipments, agents and branch-level activities.

## 2.3 Brainstorming

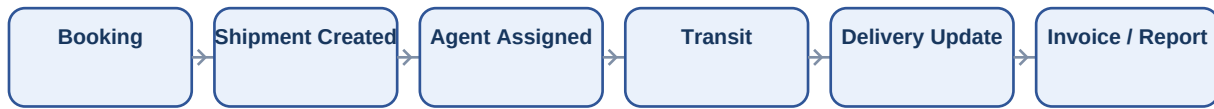
| Idea                | Strength                    | Limitation                          |
|---------------------|-----------------------------|-------------------------------------|
| Manual Register     | Easy to start               | Hard to search and update           |
| Spreadsheet         | Simple digital storage      | Limited automation and coordination |
| Separate Web Portal | Can be customized           | Requires additional infrastructure  |
| Salesforce CRM      | Cloud based with automation | Needs configuration and learning    |

## 2.4 Selected Approach

For this project, Salesforce CRM is used as the implementation platform because the reference project is built around Salesforce features such as custom objects, Flow automation, security settings and reports. The student implementation can therefore demonstrate an end-to-end business application without building every infrastructure component from scratch.

## 3. REQUIREMENT ANALYSIS

### 3.1 Customer Journey



The journey begins when shipment information is recorded and continues through assignment, movement, delivery status updates and business reporting.

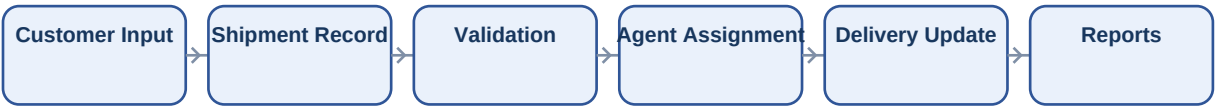
### 3.2 Functional Requirements

| ID    | Requirement                                                                 |
|-------|-----------------------------------------------------------------------------|
| FR-01 | Create and maintain customer records.                                       |
| FR-02 | Create shipment records with source, destination, weight, price and status. |
| FR-03 | Associate a shipment with a delivery agent.                                 |
| FR-04 | Record delivery status updates.                                             |
| FR-05 | Maintain branch and invoice information.                                    |
| FR-06 | Generate operational reports and dashboards.                                |
| FR-07 | Apply validation rules to important fields.                                 |

### 3.3 Non-Functional Requirements

| Area         | Requirement                                                      |
|--------------|------------------------------------------------------------------|
| Usability    | A clear Salesforce Lightning interface should be used.           |
| Security     | Access should be separated according to user responsibilities.   |
| Reliability  | Automation should execute consistently for valid records.        |
| Performance  | Common records and tracking information should load efficiently. |
| Availability | The cloud platform should support access when required.          |
| Scalability  | The design should allow additional branches and records.         |

### 3.4 Data Flow



### 3.5 Technology Stack

| Layer           | Salesforce Feature              |
|-----------------|---------------------------------|
| User Interface  | Salesforce Lightning            |
| Business Logic  | Salesforce Flow                 |
| Data Storage    | Custom Salesforce Objects       |
| Data Validation | Validation Rules                |
| Security        | Profiles, Roles and Permissions |
| Analytics       | Reports and Dashboards          |

## 4. PROJECT DESIGN

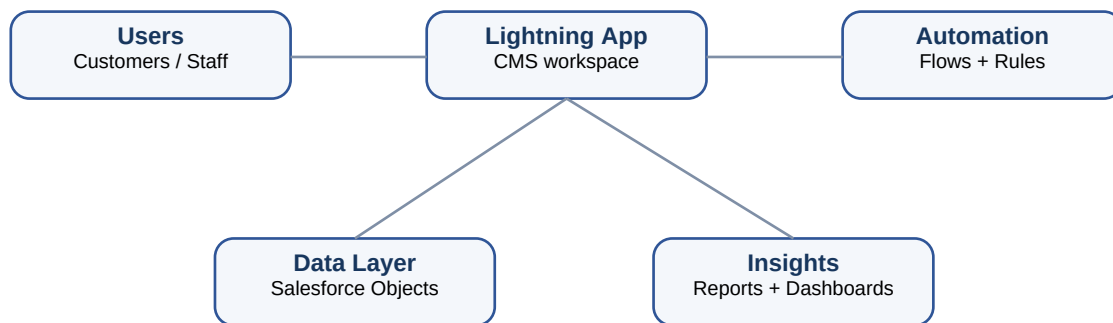
### 4.1 Problem–Solution Fit

| Business Issue                 | CMS Response                 |
|--------------------------------|------------------------------|
| Scattered shipment information | Centralized Shipment records |
| Incorrect data entries         | Validation rules             |
| Manual status propagation      | Record-triggered Flow        |
| Unclear user permissions       | Profiles and roles           |
| Difficult performance review   | Reports and dashboards       |

### 4.2 Proposed Solution

CMS is organized around connected Salesforce objects. Customer information is linked to shipments; shipments can reference delivery agents; agents are associated with branches; delivery updates belong to shipments; and invoices can reference shipment records.

### 4.3 Solution Architecture



### 4.4 Main Components

- Customer – stores customer identity and contact details.
- Shipment – stores tracking, address, weight, price, status and expected date.
- Delivery Agent – stores agent contact, branch and working status.
- Branch – represents courier office information.
- Delivery Update – stores shipment status changes and remarks.
- Invoice – stores billing and payment information.
- Flows – automate selected record updates.
- Reports and Dashboards – present operational information.



## 5. PROJECT PLANNING & SCHEDULING

### 5.1 Planning Approach

The project follows a simple Agile-style approach in which the work is divided into manageable sprints. Each sprint focuses on a functional area of the application.

| Sprint | Epic           | Example User Story                                       | Points | Priority |
|--------|----------------|----------------------------------------------------------|--------|----------|
| 1      | Platform Setup | Create and configure the Salesforce project environment. | 3      | High     |
| 2      | Data Modeling  | Create the core CMS objects and relationships.           | 5      | High     |
| 3      | Automation     | Add validations and shipment status automation.          | 5      | High     |
| 4      | Security       | Configure profiles and role hierarchy.                   | 5      | High     |
| 5      | Reporting      | Build operational reports.                               | 4      | High     |
| 6      | Dashboard      | Create dashboard components for monitoring.              | 4      | Medium   |

### 5.2 Project Tracker

| Sprint   | Planned Focus | Expected Output                                       |
|----------|---------------|-------------------------------------------------------|
| Sprint 1 | Environment   | Developer org and initial configuration               |
| Sprint 2 | Objects       | Customer, Shipment, Agent, Branch, Update and Invoice |
| Sprint 3 | Rules & Flow  | Validation rules and status automation                |
| Sprint 4 | Security      | Profiles, roles and user access                       |
| Sprint 5 | Analytics     | Reports                                               |
| Sprint 6 | Visualization | Dashboard                                             |

### 5.3 Estimation Note

Story points are used as relative estimates. A larger point value represents work that generally needs more configuration or testing. The exact duration can be adjusted according to the student team's schedule and Salesforce environment.

## 6. PROJECT DEVELOPMENT PHASE

### 6.1 Salesforce Developer Account

Salesforce provides a cloud environment in which the CMS configuration can be created. A developer account is used for learning and project development.

Typical setup sequence:

- Open the Salesforce Developer sign-up page.
- Enter the requested personal and organization information.
- Activate the account through the verification email.
- Set the password and complete the initial login.
- Open Setup to begin configuring the application.

### 6.2 Understanding Salesforce Objects

A Salesforce object can be understood as a structured data container. Fields store individual values and records represent individual entries.

| Object             | Purpose                                  |
|--------------------|------------------------------------------|
| Customer__c        | Customer profile and contact information |
| Shipment__c        | Parcel and tracking information          |
| Delivery_Agent__c  | Delivery personnel details               |
| Branch__c          | Courier branch information               |
| Delivery_Update__c | Shipment status history                  |
| Invoice__c         | Billing and payment information          |

### 6.3 Object Relationships

| Child           | Field             | Parent         | Relationship  |
|-----------------|-------------------|----------------|---------------|
| Shipment        | Customer__c       | Customer       | Lookup        |
| Shipment        | Assigned_Agent__c | Delivery Agent | Lookup        |
| Delivery Agent  | Branch__c         | Branch         | Lookup        |
| Delivery Update | Shipment__c       | Shipment       | Master-Detail |
| Invoice         | Shipment__c       | Shipment       | Lookup        |

## 6.4 Customer Fields

| Field         | API Name         | Type      | Required |
|---------------|------------------|-----------|----------|
| Customer Name | Name             | Text      | Yes      |
| Email         | Email__c         | Email     | Yes      |
| Phone         | Phone__c         | Phone     | Yes      |
| Address       | Address__c       | Text Area | Yes      |
| Customer Type | Customer_Type__c | Picklist  | Yes      |
| City          | City__c          | Text      | No       |
| State         | State__c         | Text      | No       |
| Pincode       | Pincode__c       | Text      | No       |
| Is Active     | Is_Active__c     | Checkbox  | No       |

## 6.5 Shipment Fields

| Field               | API Name               | Type        | Required |
|---------------------|------------------------|-------------|----------|
| Tracking Number     | Name                   | Auto Number | Yes      |
| Customer            | Customer__c            | Lookup      | Yes      |
| Source Address      | Source_Address__c      | Text Area   | Yes      |
| Destination Address | Destination_Address__c | Text Area   | Yes      |
| Weight              | Weight__c              | Number      | Yes      |
| Price               | Price__c               | Currency    | Yes      |
| Status              | Status__c              | Picklist    | Yes      |
| Expected Date       | Expected_Date__c       | Date        | Yes      |
| Assigned Agent      | Assigned_Agent__c      | Lookup      | No       |
| Shipment Type       | Shipment_Type__c       | Picklist    | Yes      |

## 6.6 Delivery Agent, Branch, Update and Invoice Fields

| Object          | Important Fields                                               |
|-----------------|----------------------------------------------------------------|
| Delivery Agent  | Agent Name, Email, Phone, Branch, Status, Joining Date         |
| Branch          | Branch Name, Branch Code, City, State, Manager Name            |
| Delivery Update | Update Number, Shipment, Status, Update Date, Remarks          |
| Invoice         | Invoice Number, Shipment, Amount, Invoice Date, Payment Status |

## 6.7 Creating Custom Objects – Activity Guide

For each custom object, open Setup → Object Manager → Create → Custom Object. Enter the label, plural label and record name type. Enable the required optional features and save the object.

| Object          | Suggested Record Name         |
|-----------------|-------------------------------|
| Customer        | Customer Name (Text)          |
| Shipment        | Tracking Number (Auto Number) |
| Delivery Agent  | Agent Name (Text)             |
| Branch          | Branch Name (Text)            |
| Delivery Update | Update Number (Auto Number)   |
| Invoice         | Invoice Number (Auto Number)  |

## 6.8 Creating Tabs

Custom tabs make the objects easier to access from the Lightning application. Create tabs for Customer, Shipment, Delivery Agent, Branch, Delivery Update and Invoice, selecting an appropriate tab style for each.

## 6.9 Creating Fields

For each object, use Fields & Relationships → New, choose the required data type, provide the field label, configure required settings or picklist values, and save. Relationship fields should be created using Lookup or Master-Detail according to the data model.

## 6.10 Validation Rules

Validation rules check whether entered information meets a defined condition. When the condition evaluates as true, Salesforce displays an error and prevents the record from being saved.

| Object         | Rule                 | Formula / Condition          | Message                                       |
|----------------|----------------------|------------------------------|-----------------------------------------------|
| Customer       | Email required       | ISBLANK(Email__c)            | Email is required.                            |
| Customer       | Phone length         | LEN(Phone__c) <> 10          | Phone number must be 10 digits.               |
| Customer       | Pincode length       | LEN(Pincode__c) <> 6         | Pincode must be 6 digits.                     |
| Shipment       | Positive weight      | Weight__c <= 0               | Weight must be greater than zero.             |
| Shipment       | Positive price       | Price__c <= 0                | Price must be greater than zero.              |
| Shipment       | Future expected date | Expected_Date__c < TODAY()   | Expected delivery date cannot be in the past. |
| Delivery Agent | Email format check   | NOT(CONTAINS(Email__c, "@")) | Enter a valid email address.                  |
| Branch         | Branch code required | ISBLANK(Branch_Code__c)      | Branch Code is mandatory.                     |

## 6.11 Delayed Shipment Remark Rule

A delivery update marked as Delayed should contain a remark. A suitable condition is:

**AND( ISPICKVAL(Status\_\_c, "Delayed"), ISBLANK(Remarks\_\_c) )**

Error message: **Remarks are required when status is Delayed.**

## 6.12 Page Layouts

Page layouts control how fields and related information are arranged on a Salesforce record page. The CMS can group fields into sections so users can enter information in a logical order.

| Object          | Suggested Sections                                          |
|-----------------|-------------------------------------------------------------|
| Customer        | Customer Information; Address Details                       |
| Shipment        | Shipment Details; Address Information; Delivery Information |
| Delivery Agent  | Agent Details; Additional Information                       |
| Branch          | Branch Information                                          |
| Delivery Update | Update Information                                          |
| Invoice         | Invoice Details                                             |

## 6.13 Record-Triggered Flow

A record-triggered Flow can update the shipment status whenever a new Delivery Update record is created.



Configuration summary:

- Create a new Record-Triggered Flow.
- Object: Delivery\_Update\_\_c.
- Trigger: record is created.
- Use Update Records to find Shipment\_\_c where Id equals the Delivery Update's Shipment\_\_c.
- Set Shipment\_\_c.Status\_\_c to the Delivery Update Status\_\_c.
- Save the flow and activate it.

## 6.14 Lightning Application

Create a Lightning App named **Courier Management System**. Add the main CMS tabs so staff can move between customers, shipments, agents, branches, delivery updates, invoices, reports and dashboards.

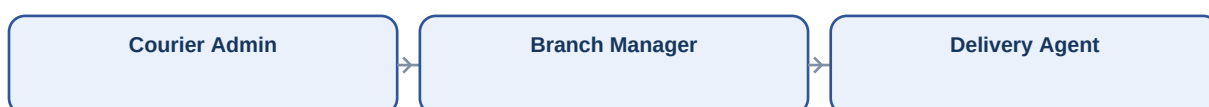
Suggested navigation items:

- Customers
- Shipments
- Delivery Agents
- Branches
- Delivery Updates
- Invoices
- Reports
- Dashboards

## 6.15 Profiles

| Profile          | Main Purpose                                      |
|------------------|---------------------------------------------------|
| Courier Admin    | Full configuration and administrative control     |
| Branch Manager   | Manage branch-level shipment and agent operations |
| Delivery Agent   | Update assigned shipment delivery information     |
| Customer Support | Handle customer and tracking enquiries            |

## 6.16 Role Hierarchy



Customer Support can be positioned under the Branch Manager when a separate support role is required. The hierarchy is used to organize reporting relationships and data visibility.

## 6.17 Users

A Salesforce user is an individual account that can log in to the organization. User access is determined by settings such as license, profile and role.

| User Type        | Role             | Profile          |
|------------------|------------------|------------------|
| Courier Admin    | Courier Admin    | Courier Admin    |
| Branch Manager   | Branch Manager   | Branch Manager   |
| Delivery Agent   | Delivery Agent   | Delivery Agent   |
| Customer Support | Customer Support | Customer Support |

When creating a user, provide a unique username, valid email, role and profile. The available licenses depend on the Salesforce environment.



## 6.18 Reports

Reports convert stored records into useful summaries. The following reports are suitable for the CMS:

| Report                  | Main Fields / Grouping                                           |
|-------------------------|------------------------------------------------------------------|
| Shipment Status Summary | Tracking Number, Customer, Status, Expected Date, Assigned Agent |
| Agent Performance       | Agent Name, Branch, Shipment Count, Status                       |
| Revenue by Branch       | Tracking Number, Customer, Price, Invoice Status, Branch, Month  |
| Customer Summary        | Customer details and related shipment information                |

## 6.19 Dashboard

A CMS dashboard can contain several components that provide a quick visual overview.

| Component                | Source                   | Visualization |
|--------------------------|--------------------------|---------------|
| Shipment Status Overview | Shipment Status Summary  | Donut Chart   |
| Agent Performance        | Agent Performance Report | Bar Chart     |
| Revenue by Branch        | Revenue Report           | Column Chart  |

Recommended dashboard name: **Courier Management Dashboard**.

## 7. FUNCTIONAL & PERFORMANCE TESTING

Testing is used to confirm that records, validation rules, automation, access settings and reports behave as intended.

| Test ID | Scenario                      | Expected Result                              |
|---------|-------------------------------|----------------------------------------------|
| T01     | Create a valid customer       | Customer record saves successfully           |
| T02     | Enter an invalid phone length | Validation message appears                   |
| T03     | Enter a non-six-digit pincode | Pincode validation prevents save             |
| T04     | Enter zero shipment weight    | Weight validation prevents save              |
| T05     | Enter a past expected date    | Date validation prevents save                |
| T06     | Create a Delivery Update      | Related shipment status is updated by Flow   |
| T07     | Create report                 | Correct shipment data is displayed           |
| T08     | Open dashboard                | Charts display the configured report data    |
| T09     | Login as restricted user      | Only permitted objects/actions are available |

### 7.1 Performance Review

For this academic project, performance can be reviewed by checking whether common records open correctly, Flow execution completes without errors, and reports return the expected records. Large-scale performance measurements are not established by the reference document and should only be reported if the project team actually performs them.

## 8. ADVANTAGES & LIMITATIONS

### 8.1 Advantages

- Centralized storage of courier information.
- Reduced dependence on repeated manual status updates.
- Validation helps prevent selected data-entry errors.
- Flow automation can synchronize shipment status.
- Reports and dashboards provide operational visibility.
- Profiles and roles support controlled access.

### 8.2 Limitations

- The solution depends on Salesforce configuration and availability.
- Initial setup requires time and platform knowledge.
- Actual SMS, GPS and mobile features require additional implementation.
- Permissions and licenses must be configured correctly for users.

### 8.3 Risk Considerations

| Risk                         | Possible Response                                |
|------------------------------|--------------------------------------------------|
| Incorrect configuration      | Test each object, rule and flow before final use |
| Duplicate or incomplete data | Use validation and clear field requirements      |
| Unauthorized access          | Review profiles, roles and permissions           |
| Automation error             | Use Flow debugging and controlled test records   |
| Report mismatch              | Check filters, fields and source report types    |

## 9. CONCLUSION

The Courier Management System demonstrates how Salesforce can be configured to support courier operations through structured data, relationships, validation, automation, security and analytics. The design connects customers, shipments, delivery agents, branches, updates and invoices into a common application.

The project also demonstrates the complete development cycle from problem identification and requirements to design, configuration, testing and reporting. The resulting system provides a practical academic example of a cloud-based business application.

## 10. FUTURE SCOPE

- Mobile-friendly shipment tracking application.
- GPS-based delivery location integration.
- SMS or email notifications for shipment events.
- Additional branch-level analytics.
- AI-assisted delivery route or workload optimization.
- Customer self-service tracking interface.

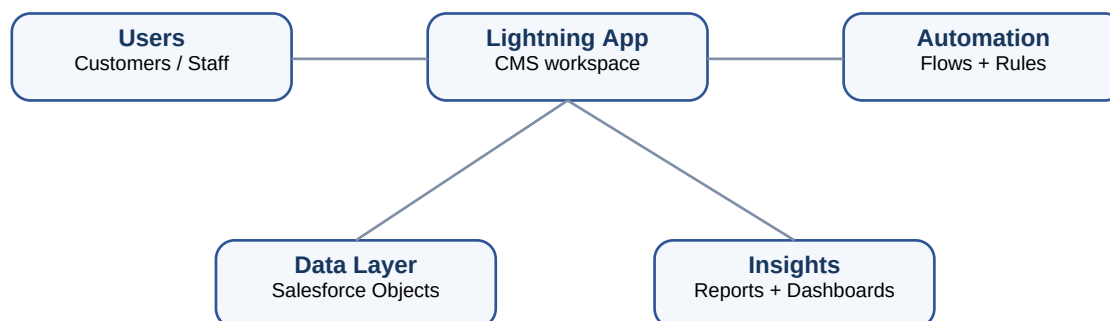
Future features should be treated as proposed extensions; they are not claimed as implemented in this document.

# 11. APPENDIX – PROJECT VISUALS

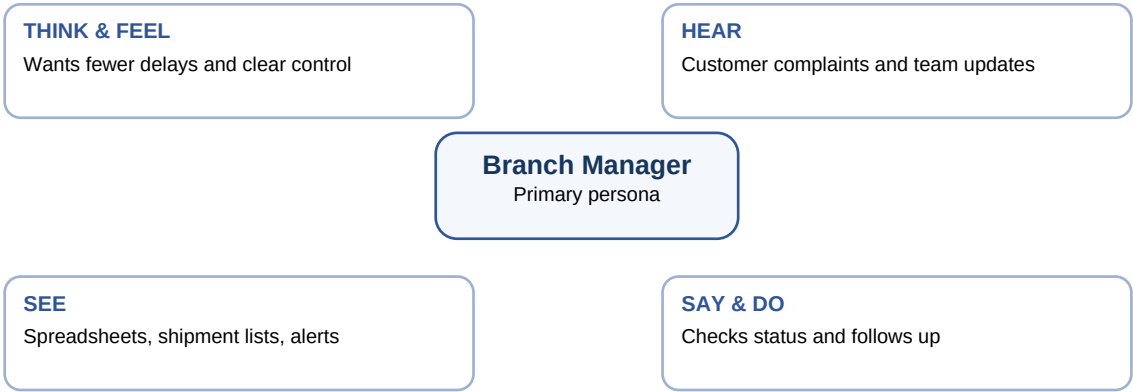
## 11.1 End-to-End Process Visual



## 11.2 CMS Architecture Visual



### 11.3 User Persona Visual



### 11.4 Data Relationship Summary

| Customer       | → | Shipment        | One customer can be linked with shipment records |
|----------------|---|-----------------|--------------------------------------------------|
| Shipment       | → | Delivery Agent  | A shipment can be assigned to an agent           |
| Delivery Agent | → | Branch          | Agent belongs to a branch                        |
| Shipment       | → | Delivery Update | Updates record shipment progress                 |
| Shipment       | → | Invoice         | Billing record references shipment               |

## 11.5 Suggested Salesforce Screenshots for Final Submission

The following screenshots can be inserted from your own Salesforce org after completing the configuration. Using your own screenshots will make the submission clearly reflect your implementation.

- Salesforce Setup / Object Manager showing CMS objects.
- Customer object fields and a sample customer record.
- Shipment record showing tracking, status and assigned agent.
- Validation Rule screen for Weight or Pincode.
- Flow Builder screen showing the shipment status automation.
- Lightning App navigation showing CMS modules.
- Profile or role configuration.
- Shipment Status report.
- Agent Performance report.
- Courier Management Dashboard.

## 11.6 Student Submission Note

Prepared by: **YUSRA MEHEK N**

Course: BCA – Third Year

Project: Courier Management System (CMS)

Platform: Salesforce

# THANK YOU